Li Jing

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Education

Massachusetts Institute of Technology — *Cambridge, MA*

PH.D. CANDIDATE IN PHYSICS — *2014 - Present*

- Research in Deep Learning and Applied Physics.
- Coursework in CS and ML: Design and Analysis of Algorithms, Artificial Intelligence, Machine Learning, Statistical Learning Theory, Computer Vision, Natural Language Processing

Peking University — *Beijing, China*

B.S. IN PHYSICS, B.A. IN ECONOMICS(DUAL MAJOR) — *2010 - 2014*

Research in Quantum Information and Quantum Computation

Selected Honor and Award

2010 **Gold Medal**, 41th International Physics Olympiad (IPhO) — *Zagreb, Croatia*

Experience

Deep Learning and Applied Physics Group in MIT — *Cambridge, MA*

RESEARCH ASSISTANCE, ADVISOR: MARIN SOLJAČIĆ — *2016 - Present*

- Applied a variety of machine learning methods to physics, e.g. photonics inverse design.
- Optimized architecture of optical neural network chip; developed hardware level training algorithms.
- Developed efficient unitary recurrent neural network.

Center for Ultracold Atoms in MIT — *Cambridge, MA*

RESEARCH ASSISTANCE, ADVISOR: WOLFGANG KETTERLE — *2014 - 2016*

- Built high power laser tapered amplifier; designed analog PID circuits used to generally stabilize laser power.
- Developed atomic physics image software system for AMO physics.

Quantum Information Group in Chinese Academy of Sciences — *Beijing, China*

UNDERGRADUATE RESEARCHER, ADVISOR: HENG FAN — *2011 - 2014*

- Investigated optimization problems in quantum cloning and quantum cryptography.
- Developed methods to design entanglement states to reduce measurement error in quantum metrology.

Qidian Technology — *Beijing, China*

QUANTITIVE RESEARCHER AND TRADOR — *2012 - 2014*

- Collected massive data from public social media website including Linkedin, Twitter, Facebook and Weibo.
- Built natural language models to evaluate employee performances in NASDAQ high-tech companies.
- Established financial models using deep neural network to predict company performances given employee performances.
- Achieved high profit rate(20% per year)

Selected Publications

2017	**Gated Orthogonal Recurrent Units: On Learning to Forget**, **Li Jing**, Caglar Gulcehre, John Peurifoy, Yichen Shen, Max Tegmark, Marin Soljačić, Yoshua Bengio	*AAAI under review*
2017	**Tunable Efficient Unitary Neural Networks (EUNN) and their application to RNNs**, **Li Jing**, Yichen Shen, Tena Dubček, John Peurifoy, Scott Skirlo, Yann LeCun, Max Tegmark, Marin Soljačić	*ICML*
2014	**Quantum Cloning Machines and the Applications**, Heng Fan, Yi-Nan Wang, **Li Jing**, Jie-Dong Yue, Han-Duo Shi, Yong-Liang Zhang, Liang-Zhu Mu	*Phys. Rep.*

Skills

Deep Learning Frameworks	Tensorflow, PyTorch, Theano
Language	English, Mandarin
Programming	Python, C++, Matlab, Lua
Full Stack Developer	Javascript, Django, AngularJS, NodeJS, Less, PHP, Swift.